# Customer Segmentation with Clustering

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## 1. Executive Summary

This project presents a comprehensive customer segmentation analysis using **K-Means clustering** on a large dataset containing nearly **951,000 customer transaction records**. The primary objective was to identify meaningful customer groups based on behavioural, demographic, and transactional attributes.

The workflow included rigorous data cleaning, type conversions, feature engineering, outlier handling through Winsorization, and feature scaling. A refined set of **15 features**, including engineered metrics such as Recency, Frequency, Monetary value, CLV, and Customer Age, formed the basis of the clustering model.

After evaluating multiple cluster sizes using the **Elbow Method** and **Silhouette Scores**, **six clusters (K = 6)** were selected as the most suitable segmentation. The resulting clusters reveal distinct customer profiles that can support targeted marketing, loyalty strategies, and operational improvements.

## 2. Introduction

Understanding customer diversity is essential for businesses aiming to optimize engagement, retention, and profitability. Customer segmentation enables organizations to categorize customers into groups with similar characteristics, allowing for more personalized and effective strategies.

This project employs **unsupervised machine learning**, specifically **K-Means clustering**, to uncover natural groupings within the customer base. Unlike supervised learning, clustering does not rely on predefined labels; instead, it identifies patterns directly from the data. The segmentation results provide valuable insights into customer behaviour, spending patterns, loyalty, and demographics.

## 3. Dataset Overview

The dataset comprises **951K rows** and includes a mixture of numerical, categorical, and date-based fields. Key attributes include:

- **Transactional Variables:** Quantity, Total Revenue, Unit Cost, Discount, Profit, Order ID
- **Customer Attributes:** Customer ID, Customer Type, Customer Group, Customer Birthdate, Loyalty Number
- **Geographical Information:** City, State/Province, Continent, Postal Code
- **Date-Based Fields:** Order Date, Delivery Date, Days to Delivery

Many fields were initially stored as text and required conversion to appropriate formats before analysis.

## 4. Data Preparation and Cleaning

### 4.1 Data Type Standardization

A significant portion of the dataset required type corrections:

- Revenue, cost, discount, and profit values were converted to numeric types
- Order and delivery dates were transformed into datetime formats
- Categorical fields were normalized for consistency

This ensured that all features were ready for analysis and modeling.

### 4.2 Handling Missing and Duplicate Records

- Missing values were addressed through imputation or removal, depending on the context
- Duplicate records were eliminated to maintain dataset integrity

### 4.3 Feature Engineering

To capture deeper behavioural and value-based characteristics, six new features were created:

- **Customer Age** (derived from birthdate)
- **Recency** (days since last purchase)
- **Frequency** (number of purchases made)
- **Monetary Value** (total spending)
- **Customer Lifetime Value (CLV)**
- **Average Unit Cost**

These engineered features significantly enhanced the clustering model's ability to differentiate customer groups.

### 4.4 Outlier Treatment

Initial outlier detection using the **IQR method** flagged nearly **40%** of the dataset—an impractical level of data loss.

To preserve the dataset while still controlling extreme values, **Winsorization** was applied to all numeric columns. This approach capped extreme values without removing large portions of data.

### 4.5 Encoding Categorical Variables

Since K-Means requires numeric inputs:

- **Customer Type** and **Customer Group** were converted into dummy variables
- This allowed categorical information to be incorporated into the clustering process

### 4.6 Feature Scaling

All numeric and encoded features were standardized using **StandardScaler**.

Scaling ensures that all features contribute equally to the distance calculations used by K-Means.

**5. Feature Selection for Clustering**

The final clustering model used the following **15 features**:

- Total Revenue
- Profit
- Quantity
- Unit Cost
- Discount
- Days to Delivery
- Loyalty Number
- Recency
- Frequency
- Monetary Value
- CLV
- Average Unit Cost
- Customer Age
- Customer Type (dummy variables)
- Customer Group (dummy variables)

These features collectively represent customer value, behaviour, demographics, and loyalty.

**6. Exploratory Data Analysis (EDA)**

EDA revealed several important patterns:

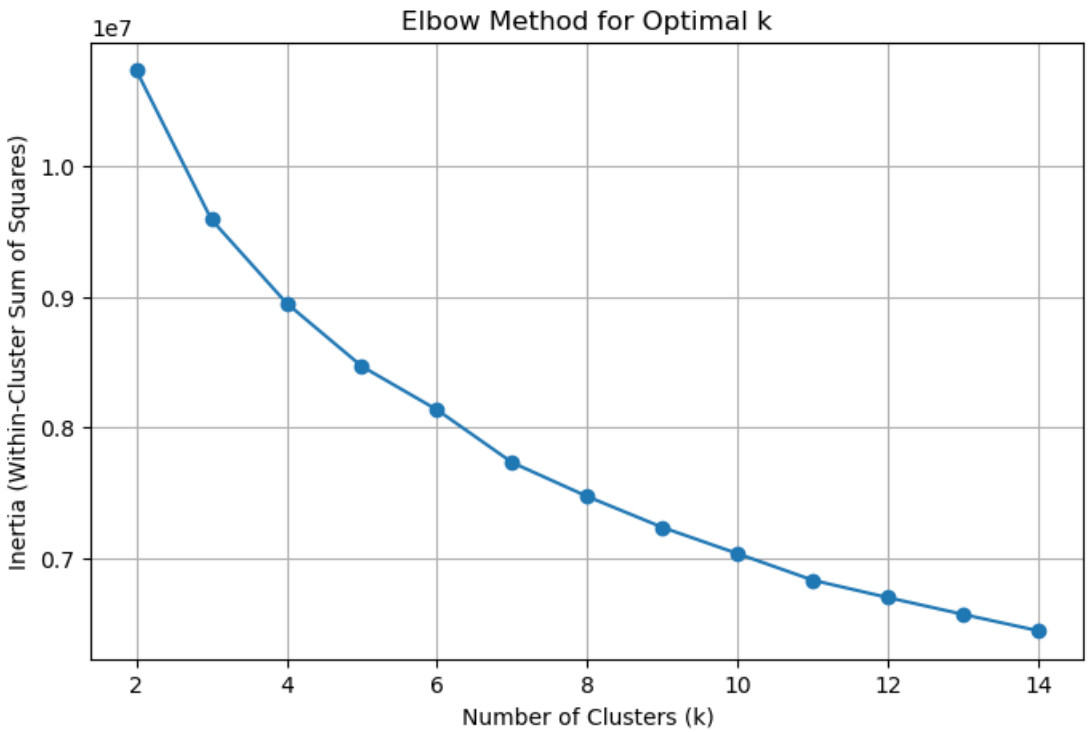
- Revenue, profit, and monetary values were heavily right-skewed
- Customer ages varied widely, indicating a diverse demographic base
- Recency and frequency distributions suggested multiple engagement levels
- Strong correlations were observed between revenue, monetary value, and CLV
- Delivery times varied significantly across regions

These insights informed decisions about feature engineering and outlier handling.

**7. Determining the Optimal Number of Clusters**

**7.1 Elbow Method**

The elbow curve showed diminishing returns in inertia reduction around **k = 6 to 8**, indicating that the ideal number of clusters lies within this range.




### 7.2 Silhouette Analysis

Silhouette scores were calculated for k = 6 to 11:

```
k=6, silhouette=0.0970
k=7, silhouette=0.0949
k=8, silhouette=0.0887
k=9, silhouette=0.0934
k=10, silhouette=0.0948
k=11, silhouette=0.0918
```

Although the scores were modest—expected for high-dimensional, large datasets—**k = 6** produced the highest score and the most interpretable segmentation.

**Final choice: K = 6 clusters**

## 8. Model Training

The K-Means model was trained using:

- **k-means++ initialization**
- 300 maximum iterations

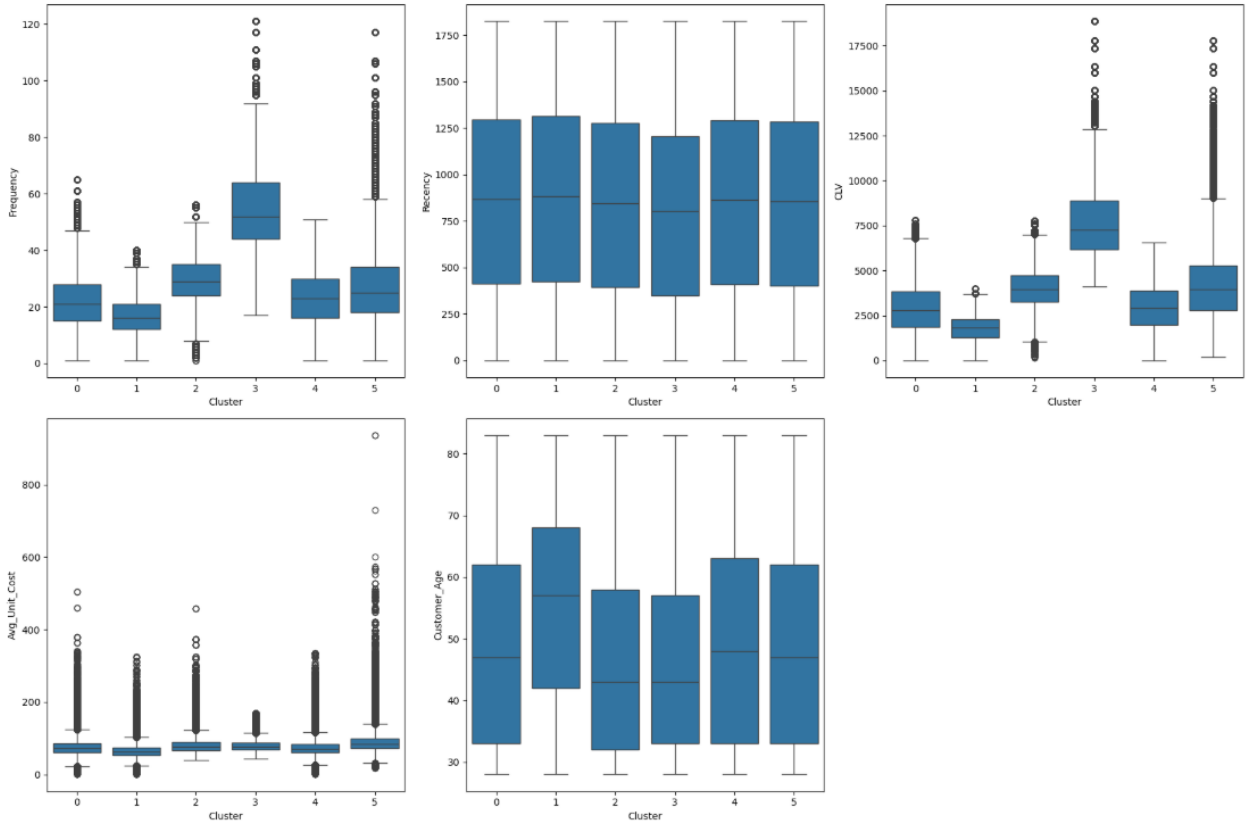
- A fixed random state for reproducibility

The algorithm converged successfully, producing stable and meaningful cluster centroids.

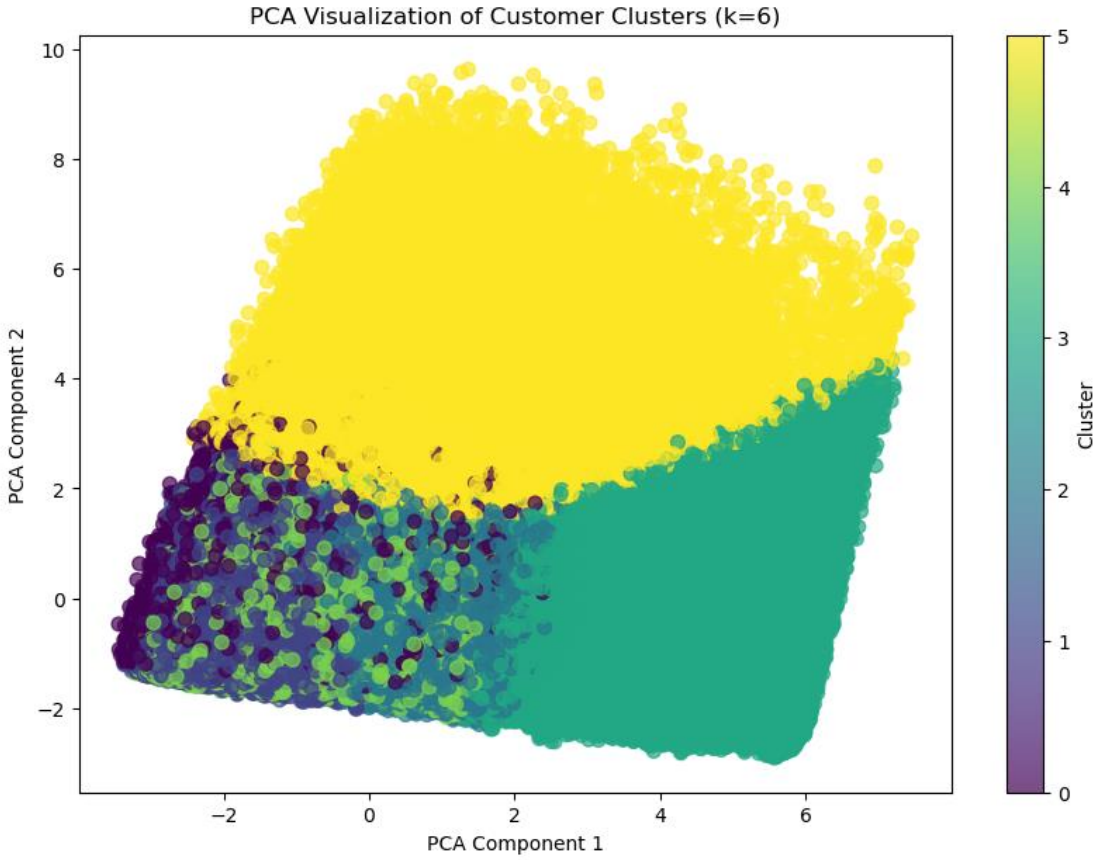
### 9. Cluster Interpretation and Insights

To understand the characteristics of each cluster:

- **Boxplots** were generated for all features across clusters




- **PCA** was used to project the high-dimensional data into a 2D space for visualization




**Key Observations**

- Some clusters represented **high-value customers** with strong profitability and frequent purchases
- Others showed **high frequency but low monetary value**, indicating cost-conscious behaviour
- Certain clusters were **highly responsive to discounts**
- Recency patterns helped distinguish **active customers from dormant ones**
- Delivery time differences suggested **geographical or operational segmentation**

The PCA visualization, while simplified, showed meaningful separation between clusters.

**10. Business Applications and Strategic Implications**

The segmentation provides several strategic opportunities:

- **Premium clusters** can be targeted with loyalty programs and exclusive offers
- **Low-engagement customers** may benefit from reactivation campaigns

- **Discount-sensitive groups** can be approached with promotional strategies
- **High-frequency but low-profit customers** may require cost-optimized engagement
- **Clusters with long delivery times** highlight potential operational inefficiencies

These insights can directly support marketing, logistics, and customer experience initiatives.

### 11. Limitations

- K-Means assumes spherical clusters, which may not fully reflect real customer behaviour
- Silhouette scores were low due to the dataset's size and dimensionality
- Dummy variables may distort distance calculations
- PCA visualization reduces 15 dimensions to 2, losing some information

### 12. Conclusion

This project successfully segmented a large customer dataset into **six meaningful clusters** using K-Means. The process involved:

- Rigorous data cleaning
- Extensive feature engineering
- Winsorization for outlier control
- Scaling and encoding
- Evaluation using elbow and silhouette methods
- Visual and statistical interpretation

The resulting clusters offer valuable insights that can guide targeted marketing, customer retention, and operational decision-making.

### 13. Future Enhancements

Potential improvements include:

- Testing alternative clustering algorithms (DBSCAN, Hierarchical Clustering, Gaussian Mixture Models)
- Using **t-SNE** or **UMAP** for richer visualizations
- Incorporating additional behavioural or demographic features
- Developing a **customer recommendation engine** using cluster labels
- Deploying the model as part of a real-time analytics pipeline